

Pritipriya Dasbehera

Bhubaneswar, India

✉ priti priya.dasbehera@niser.ac.in | 🌐 github.com/pritipriya-dasbehera

Education

National Institute of Science Education and Research

Bhubaneswar, India

Integrated Masters in Physical Science

October 2021 - Present

- Minor in Computer Sciences
- Current GPA: 9.47
- **Physics Coursework:**
Phase Transition and Renormalisation, Quantum Field Theory, Many-body physics, Non-Linear Dynamics and Chaos, Condensed Matter, Statistical Mechanics, Classical Mechanics
- **Computer Science:**
Graph Theory and Algorithm Design, Machine Learning, Complexity Theory, Parameterized algorithms
- **Relevant Lab Courses:**
Basic and Advanced Electronics, Optics, Computational, Nuclear Physics, Solid State

Work Experience

National Institute of Science Education and Research

Bhubaneswar, India

Pre-thesis project | Quantum Many-body Dynamics

Aug 2024 - Jul 2025

- Under the supervision of Dr. Anamitra Mukherjee
- Application of network theory to classify states of the exponentially large fork space
- Low rank inversion (LRI) technique for Greens function computation
- LRI application in montecarlo

MathePhysics

Remote

Summer internship | Biophysics

June 2024 - Aug 2024

- Under the supervision of Akshunna S. Dogra and Dr. William T. Redman
- Distance information from the variability of grid cell parameters

Inter-University Acceleration Centre

New Delhi, India

Summer internship | Accelerator Physics

June 2023 - July 2023

- Under the supervision of Dr. Rajesh Vikram Hariwal
- Investigations on Meta-materials for Beam Diagnostics Application

National Institute of Science Education and Research

Bhubaneswar, India

Summer internship | Computational Astronomy

May 2022 - July 2022

- Under the supervision of Dr. Nishikanta Khandai
- Study and implement brute force and Nearest Grid Point algorithm to find the Nearest Neighbors under Periodic Boundary Conditions and compare their time complexities.

University Projects

Anderson Impurity Problem

Bhubaneswar

Many-particle Physics course | Report

Aug 2024 - Nov 2024

- Under the supervision of Dr. Anamitra Mukherjee
- Study of dynamics of electrons in metal doped with impurities
- Different Born approximations and Feynman diagrams

Secure communication with Chua circuit

Bhubaneswar

NISER Open-Lab

Aug 2024 - Nov 2024

- Design and analysis of Chua circuit and study of the non-linear element constructed using op-amp as negative resistance
- Synchronization of two Chua circuits and study of its stability for different parameter values
- Use of synchronization to securely transmit digital and analog information by using chaos to encode information

Conferences and Events

International Centre for Theoretical Sciences

Bengaluru, India

Discussion meeting | Geometry, Probability, and Algorithms

12 May 2025 - 16 May 2025

Asia Pacific QEC Seminar

Remote, Youtube/Discord

Tutorial Series | Quantum Error Correction

20 Aug 2025 - 17 Dec 2025

Skills

Programming Python(Pandas, Tensorflow, PyTorch, NumPy, Scipy) Rust, Julia
Miscellaneous Cluster Computing, Tutoring
Soft Skills Critical thinking, Adaptability, Teamwork, Openness to criticism

Highlights

2024	Gold medal in Sixth semester , Academic	NISER
2024	Gold medal in Fifth semester , Academic	NISER
2024	Vice-President , Robotech Club	NISER
2023	Internship Talk , Metamaterials for accelerator physics	IUAC, Delhi, India
2023	Cleared Prelims , National Anveshika Experiment Skill Test(NAEST)	India
2022	Gold medal in Second semester , Academic	NISER
2022	Gold Medal in First semester , Academic	NISER
2022	Second position in Inquizitive Physics Quiz , IGNITE	NISER
2022	First position in Vigyan Vyakhya(Sci-Video) competition , IGNITE	NISER
2021	Inspire Scholar , Department of Science and Technology	India

Interests

Robotics Interested in bringing textbook physics and programming to life through robotics
Stargazing Occasionally find myself strolling alone enjoying the night sky
Reading Have always loved reading; thriller, mystery and bleak fiction
Sci. Comm. Regularly promote physics as a basic science to school students by showcasing interesting science experiments.

Languages

English Professional proficiency
Odia Native proficiency
Hindi Conversational proficiency